MD. MAHADI HASAN

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Highlights

- Ph.D. researcher in Electrical & Computer Engineering at **Clarkson University** (advised by Prof. Leo Yazhou Jiang), with **7.5+ years** of combined teaching, research, and industry experience in power systems, energy storage, and grid optimization.
- Currently conducting Ph.D. dissertation research on **stochastic DA BESS dispatch optimization**, alongside a separate **Avangrid-funded Benefit-Cost Analysis (BCA) project** for utility-scale battery storage.
- Developing a **Wasserstein-based Distributionally Robust Optimization (DRO)** framework for stochastic day-ahead BESS dispatch, integrating probabilistic load forecasting, an outage prediction model (OPM), and IEEE C57.91 transformer thermal-aging constraints for joint peak-shaving and resilience objectives.
- Building a **four-module Benefit-Cost Analysis (BCA) tool** (Generation, Sub-Transmission, Distribution, and Short-Lead-Time benefits) to rigorously capture non-overlapping value streams of utility-scale battery storage, with careful multi-value-stream stacking to avoid double-counting.
- Author/co-author of **19 peer-reviewed publications** (6 journal articles, 1 book chapter, 12 IEEE conference papers), cited **178+** times (Google Scholar).
- Proficient in **Python (PuLP, Gurobi, CBC, LightGBM, scikit-learn), MATLAB/Simulink, PSCAD, ETAP**, stochastic and robust optimization, and probabilistic/machine-learning forecasting.
- **7.5+ years** of prior faculty experience as **Lecturer** at the Military Institute of Science and Technology (MIST), Bangladesh, teaching power systems , energy conversion and power electronics courses to 100+ undergraduate students per term.

Academic Background

Ph.D., Electrical & Computer Engineering (Ongoing) *Aug 2025 – Present*
Clarkson University, Potsdam, NY, USA *CGPA: 4.00/4.00 Upto 3rd Semester*
Dissertation focus: BESS dispatch optimization, distributionally robust optimization, probabilistic load/outage forecasting, and distribution system planning.

M.Sc., Electrical and Electronic Engineering *2019 – 2023*
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh *CGPA: 3.83/4.00*
Thesis: *An Adaptive Battery Energy Storage System Controller for Enhancing Frequency Stability of a Low-Inertia Grid*
(Supervisor: Dr. Abdul Hasib Chowdhury)

B.Sc., Electrical, Electronic & Communication Engineering *2014 – 2017*
Military Institute of Science and Technology (MIST), Dhaka, Bangladesh *CGPA: 3.95/4.00 — 1st out of 125 students*
Thesis: *Switching Transient Analysis of an Overhead and Underground Power Distribution System*
(Supervisor: Dr. Abdul Hasib Chowdhury)

Current Ph.D. Research at Clarkson University

Project: *Stochastic Day-Ahead BESS Dispatch under Load and Outage Uncertainty.*

- Formulated a **weather-conditioned, lexicographic Wasserstein-DRO** dispatch model that hedges against both load and outage uncertainty, with a soft-slack terminal state-of-charge formulation to guarantee dispatch feasibility.
- Built a **probabilistic load forecasting pipeline** using LightGBM quantile regression and Monte Carlo scenario generation across multiple lag-resolution modes, with careful hour-aligned lag-feature construction to prevent information leakage.
- Designed a two-stage **gradient-boosted outage prediction model (OPM)** (LightGBM occurrence classifier + Negative Binomial/Tweedie severity model) with walk-forward cross-validation, producing calibrated outage probabilities to populate the Wasserstein ambiguity set.
- Integrated outage risk directly into dispatch via a **soft expected-reliability LP term**, coupling resilience and economic peak-shaving objectives in a single day-ahead optimization.
- Completed Phase I: BESS peak-shaving dispatch incorporating **IEEE C57.91 transformer thermal-aging** constraints; currently extending to multi-year stochastic/DRO analysis with endogenous battery degradation (Phase II).
- Tentative comprehensive exam title: *"Joint Load–Outage Scenario-Based Robust Stochastic Optimization of Day-Ahead BESS Dispatch for Peak Shaving and Resilience."*
- Preparing manuscripts targeting **IEEE Transactions on Smart Grid**, **IEEE Transactions on Power Systems**.

Project: *Utility-Scale Battery Storage Benefit-Cost Analysis (BCA) — Avangrid Funded Project.* Multi-stream valuation framework for utility-scale BESS investment decisions.

- Designing a **four-module BCA framework** spanning **Generation**, **Sub-Transmission**, **Distribution**, and **Short-Lead-Time** benefit categories to capture the full, non-overlapping value stack of a utility-scale battery asset.
- Building an NPV-based benefit-cost analysis tool to translate technical dispatch outcomes into utility-relevant economic and planning metrics.

Teaching Experience

- **Lecturer**, Dept. of Electrical, Electronic & Communication Engineering (EECE), Military Institute of Science and Technology (MIST), Dhaka, Bangladesh [MIST Profile] *Jan 2018 – present*
  Taught Power System Analysis, Power Plant Engineering, Power Electronics, Energy Conversion I & II, Basic Electrical Circuits I & II, and Electric Service Design; course coordinator and advisor for 60+ undergraduate students.
- **Guest Lecturer**, Dept. of CSE, Bangladesh University of Professionals (BUP), Dhaka *Feb 2024 – July 2025*
- **Guest Lecturer**, Dept. of EEE, University of Asia Pacific (UAP), Dhaka *Jan – Dec 2023*
- **Adjunct Lecturer**, Dept. of EEE, Northern University Bangladesh (NUB), Dhaka *May – Sep 2018*

Industry & Consultancy Experience

- Conducted acceptance testing of **7.5 MVA and 10 MVA 33/11 kV power transformers** and a **3 MVA 11/0.4 kV distribution transformer** as an electrical consultant for the Centre for Advisory and Testing Services (CATS), MIST.
- Conducted a **technical energy audit** at Windy Laundry Ltd. and inspected an 11/0.4 kV substation at Cotton Club Bangladesh, Gazipur, as a CATS electrical consultant.
- Led **feasibility studies** for the Tangail Economic Zone electrical infrastructure and the Jolshiri underground distribution system, and designed electrical services for multiple LGED government projects.

Peer-Reviewed Publications

Google Scholar (Cited by 178+) **19 peer-reviewed works: 6 journal articles, 1 book chapter, 12 IEEE international conference papers**

Journal Articles

- I. Jahan, P. Chowdhury, M. A. Kabir, **M. M. Hasan**, *et al.*, O. Farrok. "Simultaneous Peak Efficiency Attainment in FCHEV Powertrains: An Iterative Co-Optimization Approach." *International Journal of Hydrogen Energy*, Elsevier, 2026.
- **M. M. Hasan**, M. I. H. Emon, H. Monir, M. A. Kabir. "Impact of Inverter Control and Solar Photovoltaic Integration on Small Signal Stability Augmented with Battery: Modal Analysis Approach." *MIST International Journal of Science and Technology (MIJST)*, Vol. 13, 2025, pp. 73–84.
- M. A. Kabir, **M. M. Hasan**, T. Hossain, A. Ahnaf, H. Monir. "Sustainable Energy Transition in Bangladesh Academic Buildings: A Techno-Economic Analysis of Photovoltaic-Based Net-Zero Energy Systems." *Energy and Buildings*, Elsevier, 2024.
- S. Das, H. M. Ishraq, **M. M. Hasan**, M. A. Kabir. "Empirical Analysis of Power Quality Using UPQC with Hybrid Control Techniques." *Results in Engineering*, Elsevier, 2023.
- **M. M. Hasan**, A. H. Chowdhury. "An Improved Adaptive Hybrid Controller for Battery Energy Storage System to Enhance Frequency Stability of a Low Inertia Grid." *Journal of Energy Storage*, Elsevier, 2022.
- S. M. Hossain, **M. M. Hasan**. "Energy Management through Bio-gas Based Electricity Generation System during Load Shedding in Rural Areas." *TELKOMNIKA*, 2018.

Book Chapter

- I. Jahan, M. A. Kabir, **M. M. Hasan**, R. Islam, A. K. Azad, M. S. H. Khan, O. Farrok, M. S. Ali. "Proton Exchange Membrane Fuel Cell System Integration, Modelling, and Simulation for Energy-Efficient Electric Vehicle." In *Hydrogen Energy Conversion and Management*, Elsevier, 2023.

Selected IEEE International Conference Papers (12 total)

- **M. M. Hasan**, A. H. Chowdhury. "Droop Control Based Virtual Inertia Emulation for Stability Enhancement of a Microgrid System." *ICECE*, IEEE, 2022.
- **M. M. Hasan**, A. H. Chowdhury. "Performance Analysis of a DQ0 Controlled Grid Forming Inverter for Grid Connected Photovoltaic System." *STI*, IEEE, 2022.
- **M. M. Hasan**, M. R. Hoque, R. Islam, K. Islam. "Comparative Analysis of Harmonic Mitigation Schemes for a Three-Phase Uncontrolled Full-Wave Rectifier." *ICEPE*, IEEE, 2022.
- F. H. Sneha,**M. M. Hasan**,M. G. Mostafa. "Improved Design of a Battery Charger for Three-Wheeler Auto Rickshaws in Bangladesh with Low Harmonic SWISS Rectifier." *ACMI*, IEEE, 2021.
- **M. M. Hasan**, S. Biswas, M. A. A. Khan. "Study of Switching Transient Over-Voltage due to Transient Short-Circuit Fault in a Power Distribution Network." *ICREST*, IEEE, 2019.
- *Plus 7 additional IEEE conference papers on distributed generation, net metering, and PV penetration analysis (full list on Google Scholar).*

Awards & Recognition

- Awarded with **MIST Medal**, for securing **1st position** among 125 students in the B.Sc. Engineering program (2017)
- Awarded with **Dean's Certificate & Scholarship**, B.Sc. Engineering program (2017)
- Awarded with **MIST Commandant's Certificate and Scholarship** for being in the **top 5 students** list in the B.Sc. Engineering program (2014–2016)
- **1st Position**, Inter-University EEE Olympiad, IEEE Student Branch, MIST, 2014

Professional Service & Memberships

- Conducted two days workshop on "Advanced AutoCAD for Electrical Systems Design" at University of Asia Pacific (UAP) organized by IEEE UAP student branch & UAP EEE project club (November 2024)
- Conducted a day-long workshop on “AutoCAD for Electrical Services Design,” University of Asia Pacific, organized by the IEEE UAP Student Branch & UAP EEE Project Club (October 2023)
- Technical Program & Organizing Committee Member, *6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*, 2018 & 2024, MIST
- Graduate Student Member, **IEEE**; Member (M/44235), **Institution of Engineers, Bangladesh (IEB)**

Technical Skills

- **Optimization & Programming:** Python (PuLP, Gurobi, CBC, scikit-learn, LightGBM), MATLAB/Simulink, stochastic & distributionally robust optimization, mixed-integer/linear programming
- **Power Systems Software:** PSCAD, ETAP, AutoCAD (electrical services design)
- **Forecasting & Data Analysis:** Probabilistic load forecasting, Monte Carlo simulation, gradient-boosted models, time-series CV
- **Domain Expertise:** Power System Stability, BESS dispatch/sizing, Robust Optimization, transformer thermal aging (IEEE C57.91), grid resilience, energy economics